Agentic AI

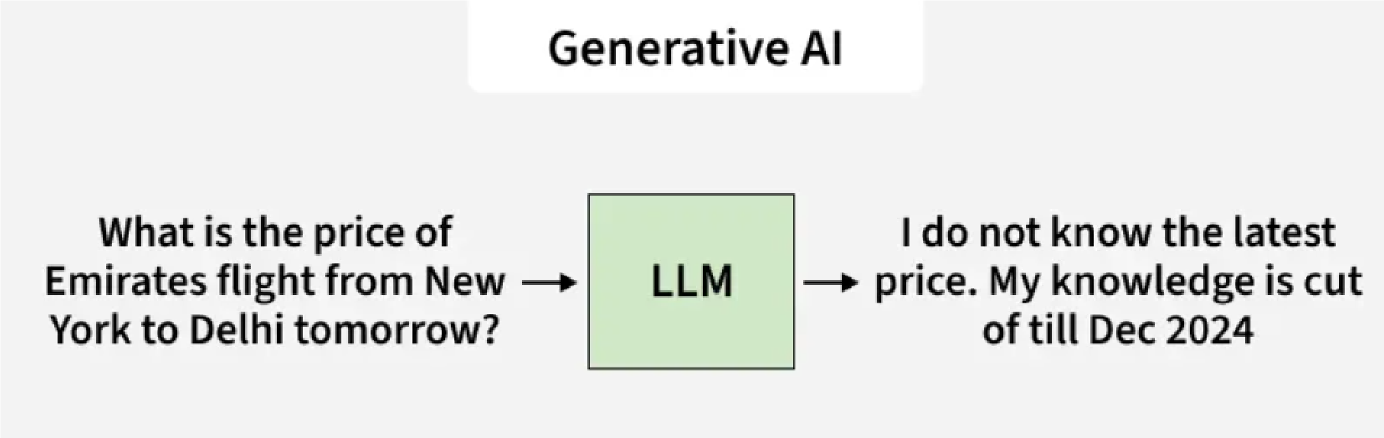
Module-1

Definition and Concept of Agentic AI, Key differences from traditional AI and generative AI, Motivation: Multi-step reasoning, autonomy, goal-driven behavior, Real-world applications and use cases

What is Generative AI?

- Refers to models that can create new content such as text, images, videos or code by learning from existing data. It doesn’t perform real actions, it simply provides responses based on available data.
- Trained on large datasets using deep learning.
- Generates creative outputs (text, design, audio, etc.).
- Works based on user prompts and learned patterns.
- Common examples: ChatGPT, DALL·E, Midjourney, Gemini.

Example: When you ask, "What is the price of Emirates flight from New York to Delhi tomorrow?“

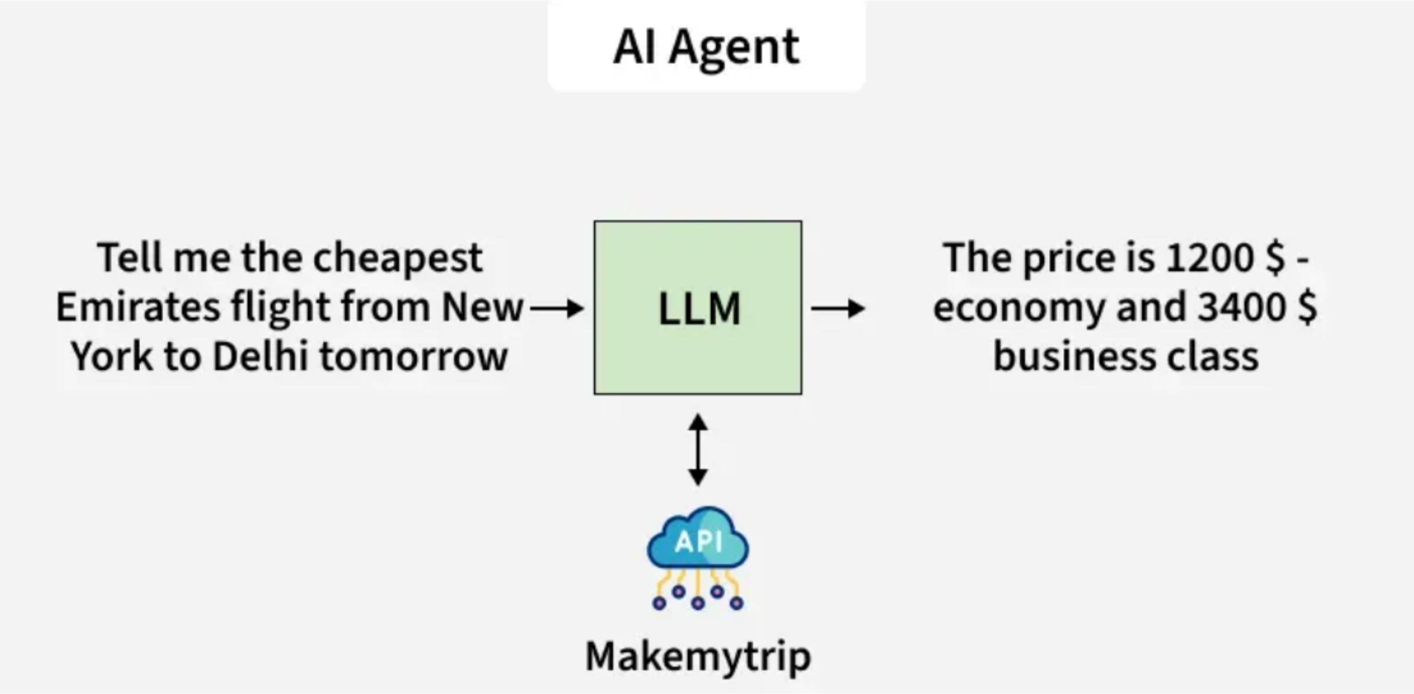


The LLM predicts and generates a possible answer based on patterns learned from its training data. However, since these models are trained on past data and do not have real-time access to flight prices or current APIs, their answers may be outdated or inaccurate. The model might respond with something like: “I do not know the latest price.”

What is AI Agent?

- AI Agents are systems designed to perform specific tasks automatically using defined instructions and external tools.
- Traditional AI Agents function within strict guidelines i.e they process inputs and respond based on preprogrammed rules, making them predictable but limited in adaptability.
- Modern AI Agents, however combine tool access and automation to handle slightly more complex tasks, though they still rely on human-defined objectives and boundaries.

Example: When you ask, “Tell me the cheapest Emirates flight from New York to Delhi tomorrow”



How it works:

1. Connects with APIs like MakeMyTrip to fetch flight data.

2. Filters results according to predefined rules such as price or class.

3. Returns the cheapest available options.

What is Agentic AI

- Agentic AI refers to autonomous AI systems that can perceive, reason, plan, and act independently to achieve complex goals with minimal human intervention.

- Agentic AI is a system or program that uses AI agents to autonomously perform tasks on behalf of a user or another system.

- Frameworks like LangChain, LlamaIndex, and AutoGen are key tools for building these systems, supporting single-agent and multi-agent setups.

- **Agentic AI** is a subset of generative AI that is centered around the orchestration and execution of agents that use LLMs as a "brain" to perform actions through tools.

- Agentic AI is focused on autonomous decision-making and action.

- Agentic AI can set goals, plan, and execute tasks with minimal human intervention.

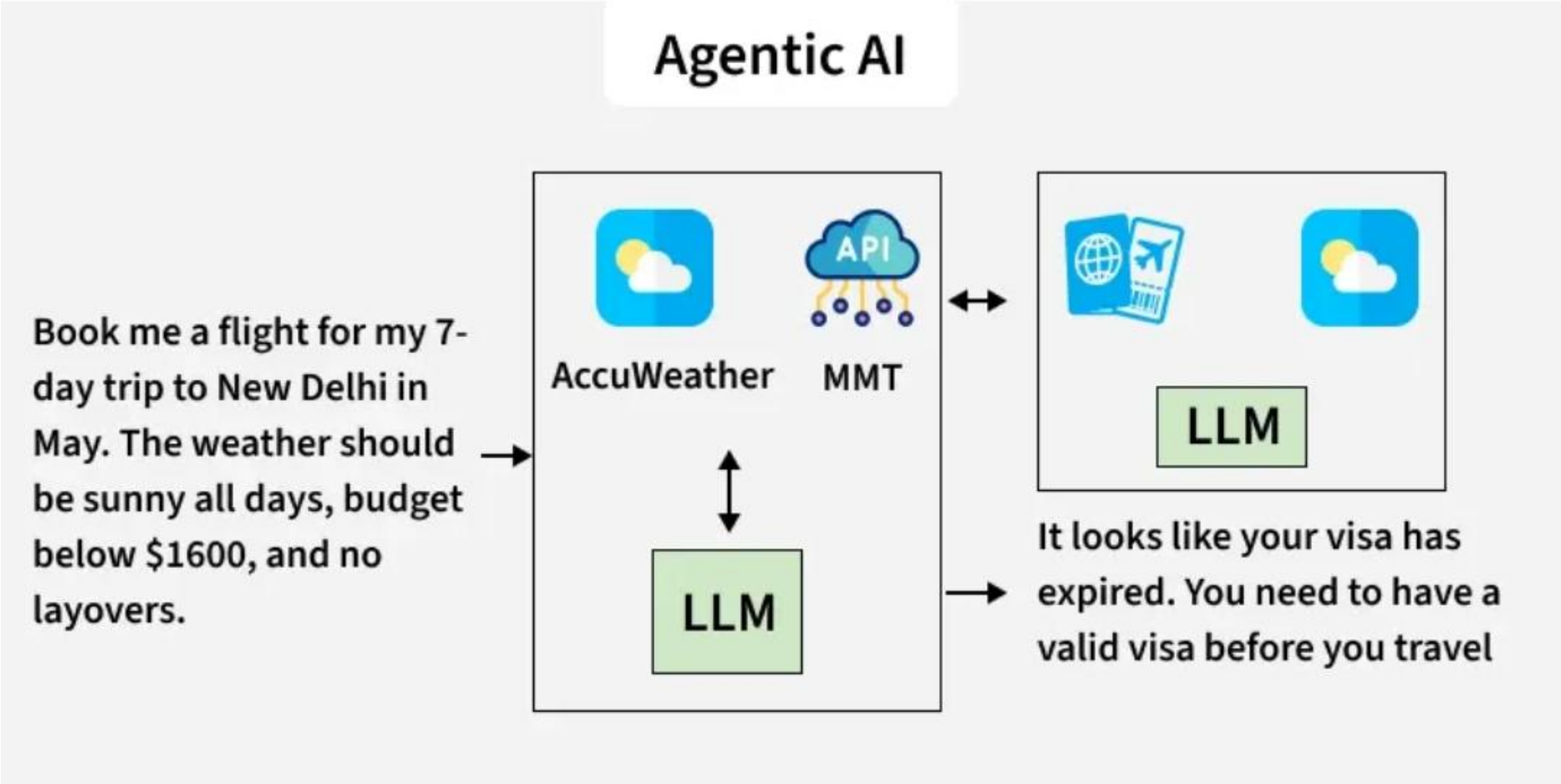
- Can accomplish a specific goal with limited supervision.

Key Characteristics

Agentic AI systems typically have:

- Autonomy: They can operate without human oversight for extended periods.
- Self-reflection: They evaluate the outcomes of their actions and adjust future behavior.
- Context-awareness: They recognize nuances in their environment and adapt accordingly.
- Tool usage: They often access APIs, apps, and data sources to complete tasks.
- Multi-step planning: They execute complex workflows rather than single commands.

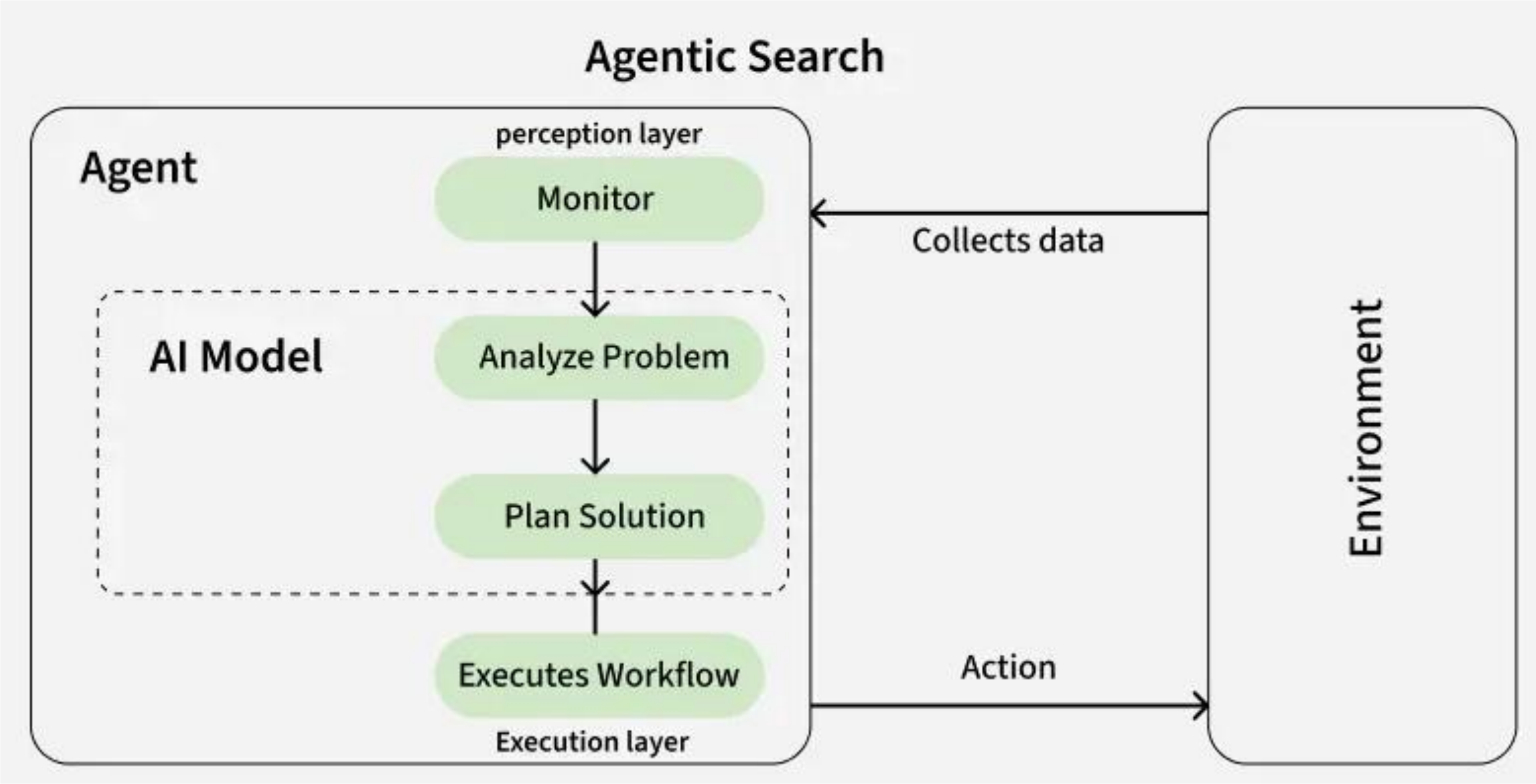
Example: When you ask, “Book me a flight for my 7-day trip to New Delhi in May. The weather should be sunny all days, budget below $1600 and no layovers.”



How Agentic AI Works

- Perception: Gathers data from the environment (user input, sensors, APIs).
- Reasoning: Interprets data, understands context, and plans actions using LLMs and other AI.
- Action: Executes tasks, potentially using tools, to achieve the goal.
- Reflection: Analyzes outcomes and self-adjusts for better future performance.

Agentic AI Architecture



Agentic AI Vs Gen AI

- Agentic AI builds on generative AI (gen AI) techniques by using large language models (LLMs) to function in dynamic environments.
- While generative AI focus on creating content based on learned patterns.
- agentic AI extends this capability by applying generative outputs toward specific goals.
- A generative AI model like OpenAI's ChatGPT might produce text, images or code, but an agentic AI system can use that generated content to complete complex tasks autonomously by calling external tools.

Agentic AI versus AI agents

- AI Agent are the building blocks of agentic AI.
- AI agents is an individual tools in a toolbox, while agentic AI is the coordinated use of those tools to build an entire house.
- While an AI agent might focus on a specific task, agentic AI employs multiple agents to handle complex workflows.
- Agentic AI acts as an overarching system that coordinates and manages these agents to achieve broader objectives.

Comparison between Gen AI vs AI Agents vs Agentic AI

Feature	Generative AI	AI Agents	Agentic AI
Primary Function	Generates new content (text, images, videos, code).	Performs specific predefined tasks automatically.	Plans, reasons and acts independently toward a goal.
Dependency	Works based on trained data.	Depends on predefined rules and APIs.	Uses reasoning, planning and feedback loops.
Tool Access	No real-time access to tools or APIs.	Limited access to tools for specific actions.	Dynamic access to multiple tools and APIs.
Learning Capability	Learns from historical data (not real-time).	No continuous learning; follows instructions.	Continuously adapts to context and outcomes.
Example Use Case	Writing articles, creating designs.	Fetching flight data or sending alerts.	Booking flights based on preferences and constraints.
Autonomy Level	Low	Medium	High
Example Tools/Models	ChatGPT, DALL·E, Midjourney	Customer service bots, automation systems	AutoGPT, Devin AI, Gemini 2.0 with planning capabilities

Types of Agentic Architectures

There are several types of agentic architectures each with its own strengths and weaknesses, suitable for different tasks and environments. Some common types include:

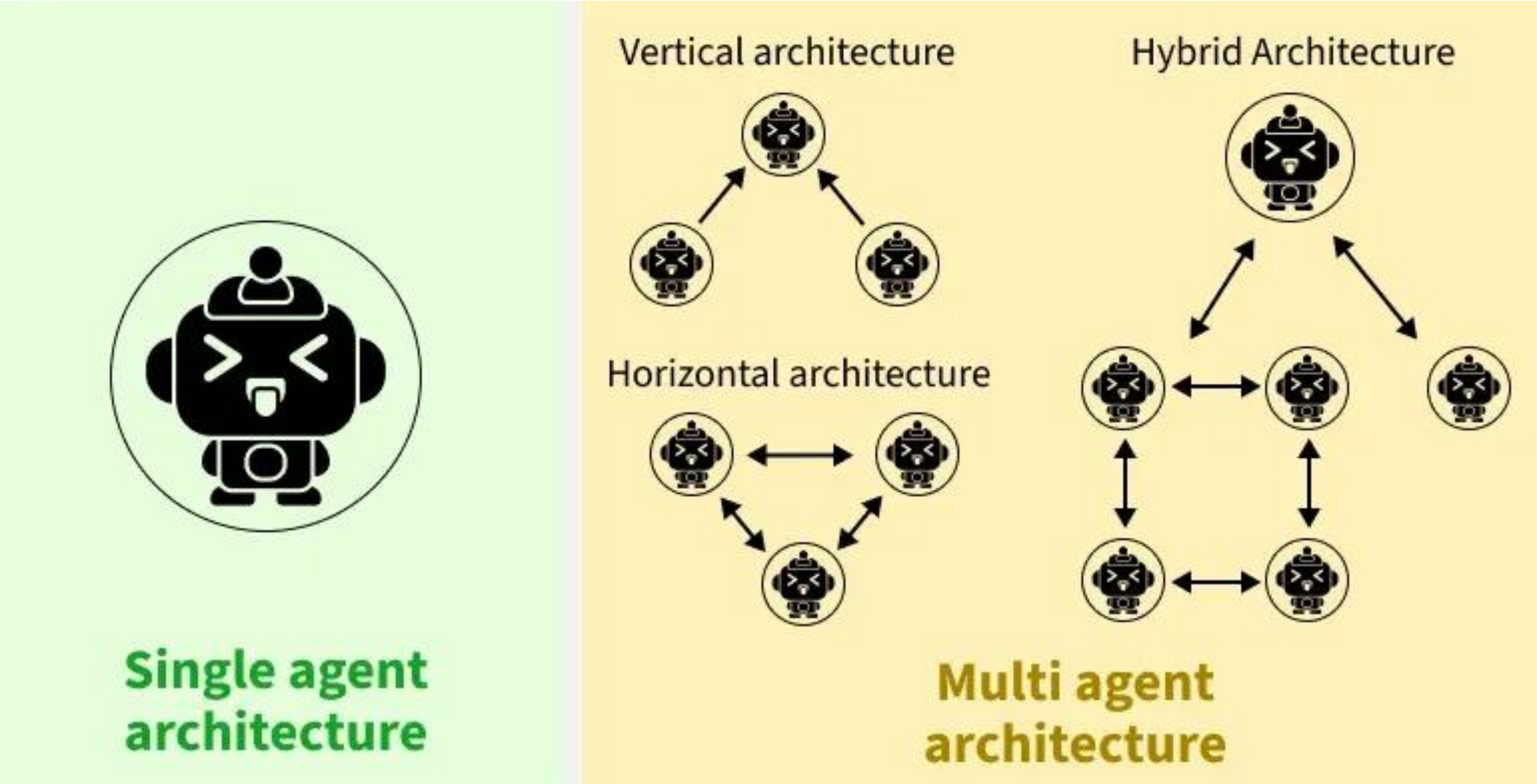
1. Single-agent architecture: A single AI system that functions independently, making decisions and taking actions without the involvement of other agents.

2. Multi-agent architecture: This architecture involves multiple AI systems interacting with each other, collaborating and coordinating their actions to achieve common goals. Its sub types are:

Vertical architecture: This approach involves agentic AI systems organized in a hierarchical structure with higher-level agents overseeing and guiding the actions of lower-level agents.

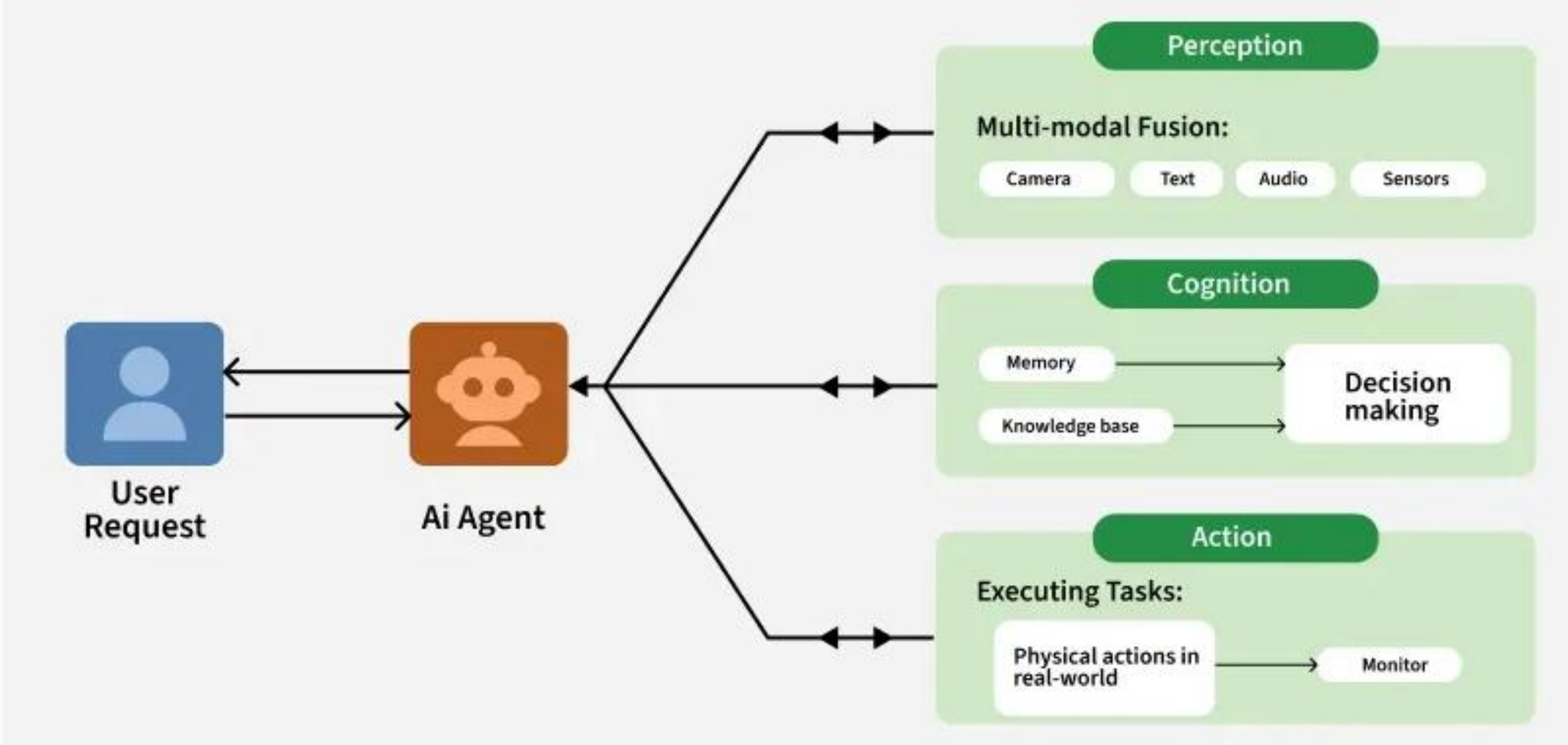
Horizontal architecture: This involves agentic AI systems operating on the same level without any hierarchical structure, communicating and coordinating their actions as needed.

Hybrid architecture: This involves a combination of different agentic architecture types and using the strengths of each to achieve optimal performance in complex environments.


Vertical architecture
Horizontal architecture
Hybrid Architecture
Single agent architecture
Multi agent architecture

Components of Single Agentic AI Architecture

The architecture of an agentic AI system is composed of several key components that work together to ensure it operates independently and effectively. These components enable the system to make decisions, adapt to new information and learn from past experi



1. Perception

- The way by which the agent collects information from its surroundings and using inputs like images, sound, text or sensor data is perception. Systems use sensors, data streams and external databases to understand their environment and recognize changes or events that need a response.
- **Sensors:** These may include cameras, microphones, motion detectors or specialized sensors designed to monitor specific aspects of the environment like temperature, location, etc.
- **Data Integration:** The AI system integrates data from multiple sources allowing for a comprehensive understanding of the situation. This can involve data from IoT devices, external APIs and historical datasets.

2. Cognitive Layer

- After understanding its environment agent must analyze the data and decide the best action. This process involves assessing the current situation, considering potential outcomes and selecting the best action based on predefined goals or objectives.

Agentic AI uses techniques such as:

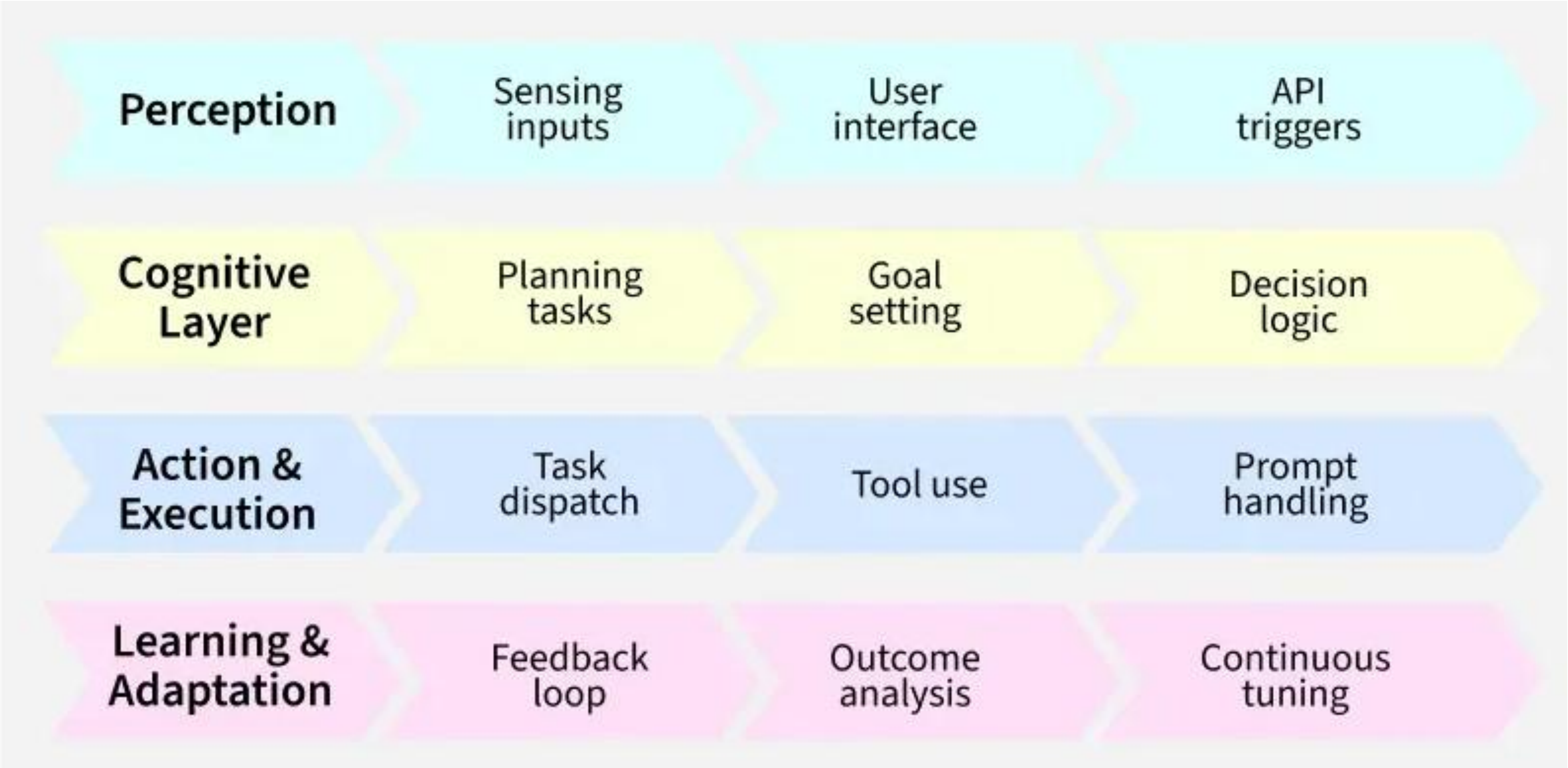
- **Rule-Based Systems:** Simple systems that follow predefined rules to make decisions.
- **Machine Learning Models:** More advanced systems that use statistical techniques to learn patterns from data and make predictions.
- **Reinforcement Learning**: Agentic AI systems often use reinforcement learning where they learn through trial and error by receiving feedback i.e rewards or penalties based on their actions.

3. Action and Execution

- The action component executes the decisions made by the agent. Once the agent processes the data and chooses an action, it takes action in the environment. This could involve sending commands to physical systems like a robotic arm or self-driving car and then handling data or communicating it with other systems.
- **Robotics:** In physical environments it can control robotic systems to perform tasks such as assembly, navigation and interaction with humans.
- **Software Automation:** In virtual environments it can control software systems to automate processes such as decision-making in business operations, customer service chatbots or IT systems management.

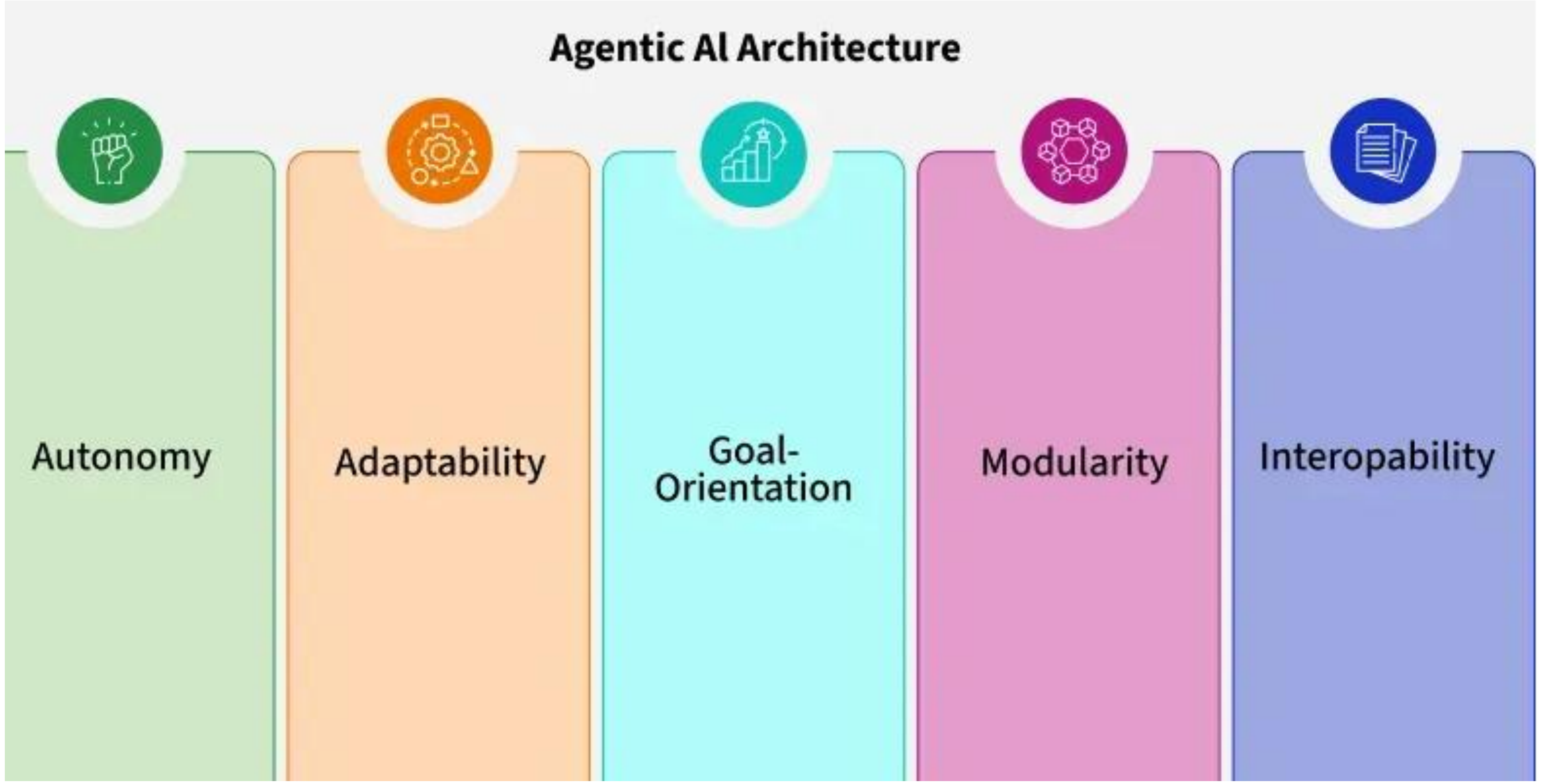
4. Learning and Adaptation

- The systems need to adapt and get better over time by learning from past experiences. This enables them to handle new situations that may not have been specifically programmed. Learning mechanisms in agentic AI can be:
- Supervised Learning: Where the agent is trained on labeled data to make predictions or classifications.
- Unsupervised Learning: Where it identifies patterns in unlabeled data without predefined categories.
- Reinforcement Learning: It learns through trial and error, improving its decision-making over time by receiving rewards or penalties based on its actions.


Perception
Sensing inputs
User interface
API triggers
Cognitive Layer
Planning tasks
Goal setting
Decision logic
Action & Execution
Task dispatch
Tool use
Prompt handling
Learning & Adaptation
Feedback loop
Outcome analysis
Continuous tuning

Principles of Agentic AI Architecture

- The principles behind the Agentic AI architecture are mentioned below:



- **Autonomy**: Agentic AI works independently within set limits and hence reducing the need for human involvement. It adapts to changing conditions while following ethical and safety guidelines.
- **Goal-Directed Behavior:** The system focuses on clear objectives, using them to guide its perception, reasoning and planning. Goals can be set by users or inferred from the context.
- **Adaptability**: Agentic systems improve over time by learning from feedback. Methods like online learning or meta-learning allow them to continuously evolve.
- **Modularity**: A modular design allows components to be developed, tested and updated independently. This enhances scalability and facilitates integration with existing systems.
- **Transparency**: To build trust, agentic AI provides understandable outputs, explaining its reasoning and actions. This is critical for applications in critical domains like healthcare or finance.

Advantages of agentic AI

Autonomous

The most important advancement of agentic systems is that they allow for autonomy to perform tasks without constant human oversight. Agentic systems can maintain long-term goals, manage multistep problem-solving tasks and track progress over time.

Proactive

- Agentic systems provide the flexibility of LLMs, which can generate responses or actions based on nuanced, context-dependent understanding, with the structured, deterministic and reliable features of traditional programming. This approach allows agents to “think” and “do” in a more human-like fashion.
- LLMs by themselves can’t directly interact with external tools or databases or set up systems to monitor and collect data in real time, but agents can. Agents can search the web, call application programming interfaces (APIs) and query databases, then use this information to make decisions and take actions

Specialized

Agents can specialize in specific tasks. Some agents are simple, performing a single repetitive task reliably. Others can use perception and draw on memory to solve more complex problems. An agentic architecture might consist of a “conductor” model powered by an LLM that oversees tasks and decisions and supervises other, simpler agents. Such architectures are ideal for sequential workflows but are vulnerable to bottlenecks. Other architectures are more horizontal, with agents working in harmony as equals in a decentralized fashion, but this architecture can be slower than a vertical hierarchy. Different AI applications demand different architectures.

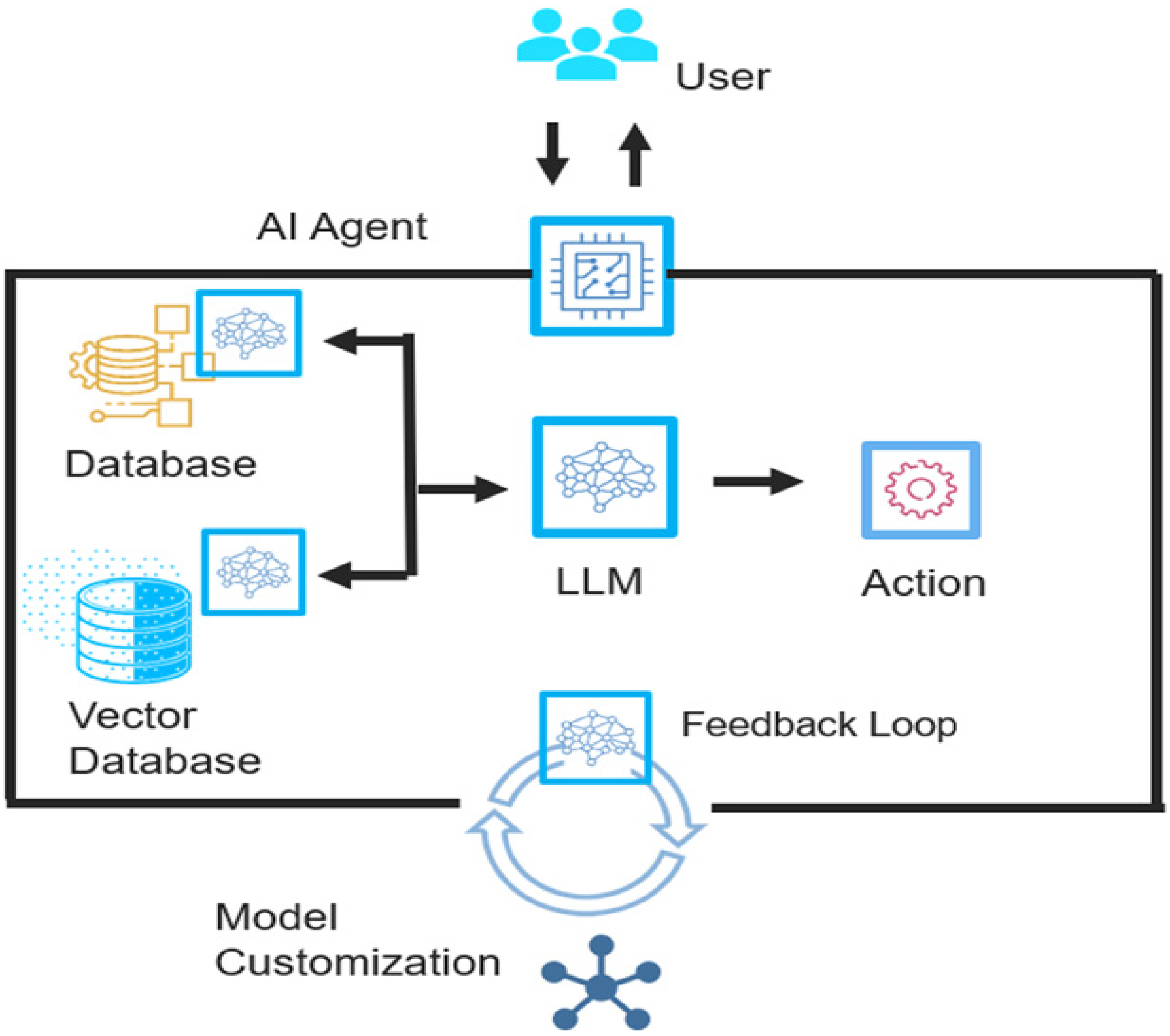
Adaptable

- Agents can learn from their experiences, take in feedback and adjust their behavior. With the right guardrails, agentic systems can improve continuously. Multiagent systems possess the scalability to eventually handle broadly scoped initiatives.

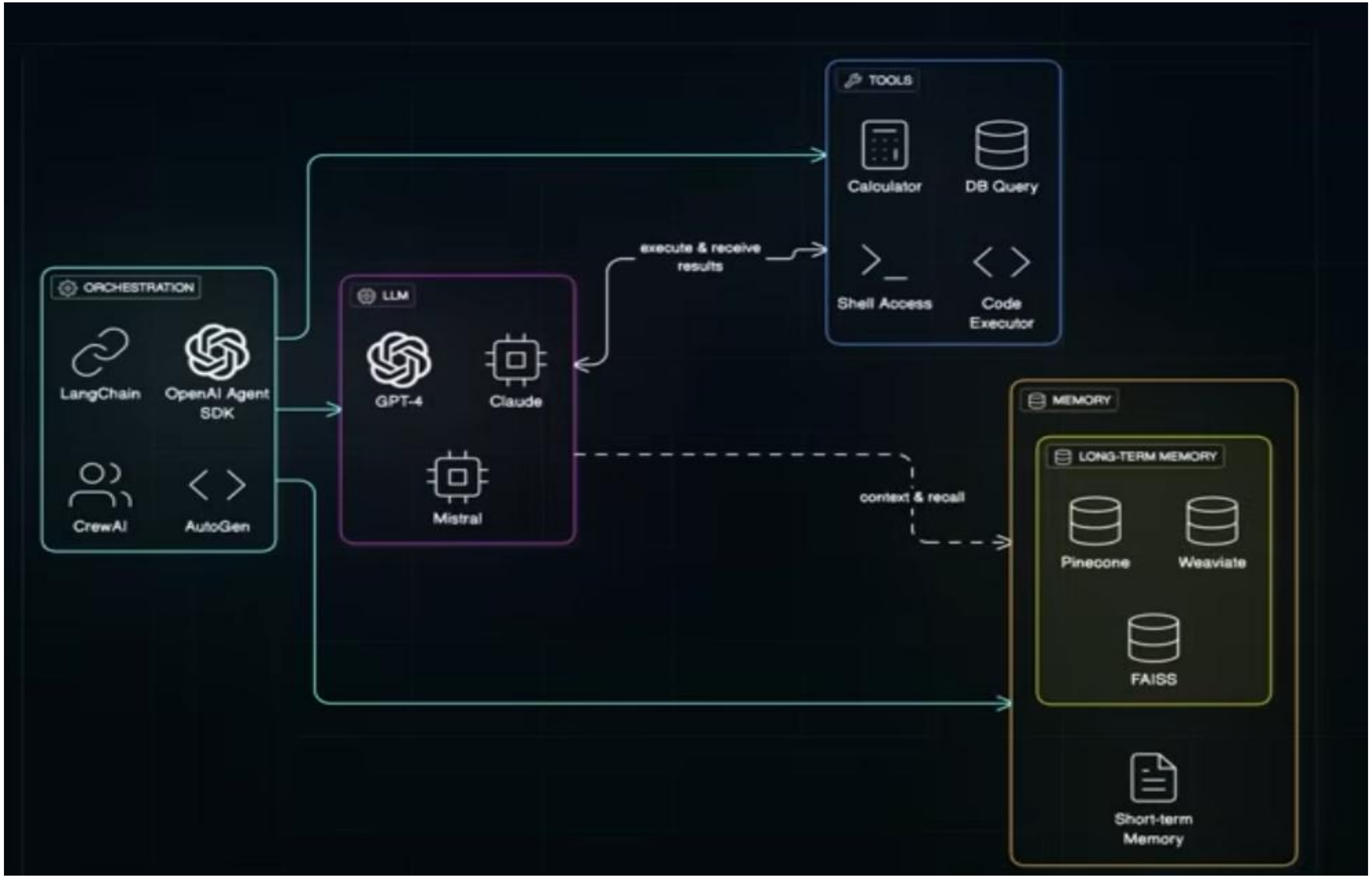
Intuitive

- Because agentic systems are powered by LLMs, users can engage with them with natural language prompts. This means that entire software interfaces—think of the many tabs, dropdowns, charts, sliders, pop-ups and other UI elements involved in the SaaS platform of one's choice—can be replaced by simple language or voice commands. Theoretically, any software user experience can now be reduced to "talking" with an agent, who can fetch the information one needs and take action based on that information. This productivity benefit can barely be overstated, when one considers the time it takes for workers to learn and master new interfaces and tools.

Agentic AI workflow



Agentic AI architecture



Orchestration Layer -LangChain, OpenAI Agent SDK, CrewAI, AutoGen

This is the **brain/controller**

- Break a task into steps
- Decide *which model* to use
- Decide *which tool* to call
- Manage multi-step workflows

LLM Layer -GPT-4, Claude, Mistral

These are the **reasoning engines**

- Understand the task
- Decide what to do next
- Generate plans, tool calls, and responses

•The orchestrator may switch between models

Tools Layer -Calculator, DB Query, Shell Access, Code Executor

•These allow the LLM to:

- Run code
- Query databases
- Execute commands
- Perform calculations

Memory Layer

- Pinecone, Weaviate, FAISS
- Long-term memory (vector databases)
- Stores
  - Past conversations
  - Knowledge embeddings
  - Learned preferences
  - Retrieved documents

- Short-term memory
  - Current session context
  - Lives only during the session

Aspect	Traditional AI	Generative AI
Approach and Techniques	Utilizes deterministic, rule-based algorithms designed for specific, structured tasks.	Employs probabilistic methods and deep learning to generate new, often unforeseen outputs from learned data.
Applications and Use Cases	- **Automation:** Used in robotic assembly lines for manufacturing. - **Diagnostic Systems:** Powers healthcare systems to diagnose diseases based on symptoms.	- **Media and Entertainment:** Creates new music, art, and scriptwriting. - **Simulations:** Generates realistic simulations for training and research in various fields.
Learning Mechanisms	Involves direct programming of specific algorithms for tasks like classification and clustering.	Utilizes advanced techniques like reinforcement learning and deep neural networks to learn from data autonomously.
Advantages	Provides predictable, reliable results and excels in environments where rules and outcomes need consistency.	Enhances creative capabilities, offering potential revolutionary applications in design, art, and data synthesis.
Limitations and Challenges	Limited to applications with clear rules and often lacks flexibility in handling new, undefined scenarios.	Raises ethical and practical concerns, such as the potential for misuse in creating realistic fake content.

Key features of agentic AI

- Decision-making: Because of the pre-defined plans and objectives these AI systems can assess situations and determine the path forward without or with minimal human input.

- Problem-solving: Agentic AI uses a four-step approach for solving issues; perceive, reason, act, and learn. These four steps start by having AI agents gather and process data. The LLM then acts as an orchestrator that analyzes perceived data to understand the situation. And is then integrated with external tools that are continuously improving and learning through feedback.

Agentic Reasoning

- Agentic reasoning is a component of AI agents that handles decision-making.
- It breaks down complex tasks into smaller, more manageable steps and then execute those steps in a logical sequence.
- This ability to plan, reason, and solve problems step-by-step is essential for tackling tasks that require more than just simple pattern recognition or information retrieval.
- Planning decomposes a task into more manageable reasoning, while tool calling helps inform an AI agent's decision through available tools.
- These tools can include application programming interfaces (APIs), external datasets and data sources such as knowledge graphs.

How do Multi-step Reasoning Agents Work?

- Multi-step reasoning agents often utilize large language models (LLMs) as their core reasoning engine.
- To enable multi-step reasoning, these agents are combined with tools and techniques that allow them to interact with external knowledge sources, perform actions, and learn from their experiences.
- Multi-step reasoning also requires that each step taken in a decision-making process be valid on its own
- capability of an AI system to solve a problem by breaking it down into a sequence of logical steps rather than producing an immediate, single-step answer.

Autonomy in Agentic AI

- Ability to make decisions and act **without constant human input**
- Emerges from **planning**, **memory**, **tool use**, and **self-monitoring loops**
- Agent can break tasks into **sub-tasks**, adapt, and work until goals are met
- Handles unexpected situations and evaluates actions dynamically
- Business value: functions like a **junior analyst**, running workflows, monitoring systems, and completing tasks end-to-end

Goal-Driven behavior

Task-solving agents are typically designed to execute specific tasks, often using predefined rules or algorithms.

goal-driven agents adapt, learn, and make decisions to achieve specific outcomes in dynamic environments.

Key Features of Goal-Driven Agents:

- Adaptive Decision-Making: Evaluates various options before acting.
- Learning-Oriented: Improves over time using feedback.
- High Complexity: Involves deep learning, reinforcement learning, etc.
- Goal-Centric Behavior: Always aligned toward achieving an end-state.

Examples:

- Self-Driving Cars
- AI Trading Bots
- Healthcare Diagnostic Tools
- Smart Assistants like Siri or Alexa

Real-World Examples of Agentic AI in Action

1. Customer Support

- AI chatbot used to rely on decision trees. Now, agentic AI systems like GPT-4-powered agents can:
- Detect customer sentiment
- Identify ticket priority
- Escalate issues automatically
- Draft and send follow-up messages

2. Healthcare Diagnostics

Agentic AI can analyze a patient's history, recommend tests, review results, and flag abnormalities..

3. Software Development

Developer-focused tools like Devin (by Cognition Labs) are agentic AI engineers who can write, test, debug, and deploy code with minimal supervision.

4. Personal Productivity

AutoGPT, BabyAGI, and Microsoft's Copilot agents are early consumer-grade examples. These tools book appointments, summarize long documents, generate reports, and make purchase decisions based on predefined goals.